National Audit of Pulmonary Hypertension Great Britain, 2018-19

Great Britain

24 October 2019

Supplementary Survival Analysis

Introduction: Overview

- The National Audit of Pulmonary Hypertension (NAPH) sets out to measure the quality of care provided to people referred to pulmonary hypertension services in Great Britain.
- Data is collected and submitted by hospital staff at pulmonary hypertension services in Great Britain. Efforts continue to be made to incorporate Northern Ireland into the Audit again.
- The Audit is delivered by NHS Digital and commissioned by NHS England and supported by NHS Scotland, NHS Wales (GIG Cymru), the Pulmonary Hypertension Association (PHA UK) and the National Pulmonary Hypertension Centres of United Kingdom and Ireland Physicians' Committee.

This supplementary report focuses on patients with a diagnosis of idiopathic, heritable, and anorexigen-induced pulmonary arterial hypertension (**IPAH**), with and without comorbidities.

Survival is split by **sex**, **age** at diagnosis and **WHO functional class**.

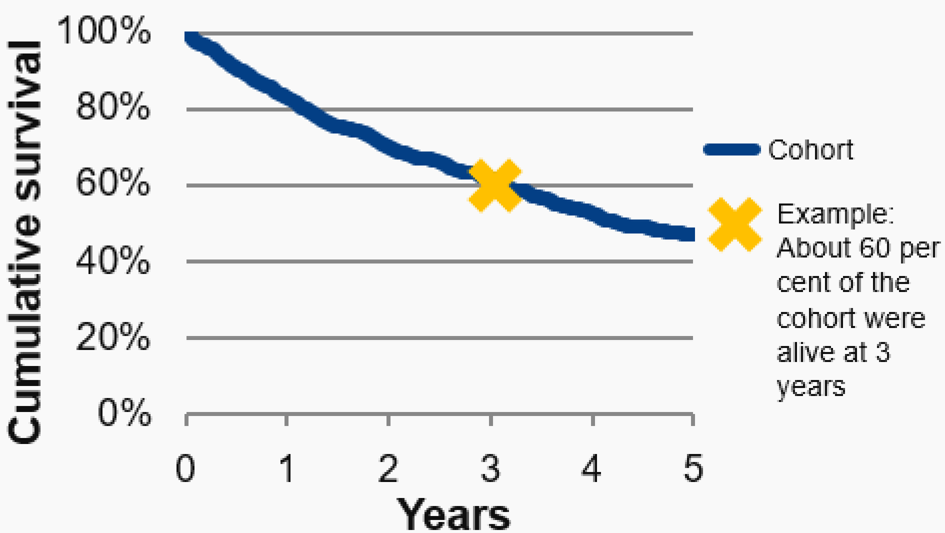
Survival curves:

Deaths in the longitudinal cohort

Survival curves: Overview

Why is this important? PH may be a life shortening condition. One of the aims of treatment is to prolong life as well as to improve its quality. Measurement of survival is used as a clinical outcome; it can be used to assess the effect of treatment on many life threatening diseases.

How is survival reported? Kaplan Meier curves are used as a way of graphically displaying an estimate of what proportion of people will still be in a group a number of days after a chosen start date. For example, it provides an estimate of the percentage of patients that will still be alive x years after they received a diagnosis (see below).



Notes: **1.** Patients at adult PH Centres aged 18 or over at latest PH diagnosis. **2.** See [Glossary: Patient cohorts].

How is survival measured?

Survival is measured from the *earliest* diagnosis date for the patient's *latest* diagnosis. 98 per cent of patients keep the same PH diagnosis.

Who is included?

Only adult patients[1] within the cohort for longitudinal analysis are included (those whose first referral was on or after 1 April 2009[2]). Patients whose diagnosis date is more than one year prior to first referral are excluded.

The survival curves are stopped if the number of patients used to calculate the cumulative survival drops below 50. This is to prevent the curve from artificially flattening due to patients whose mortality status cannot be confirmed.

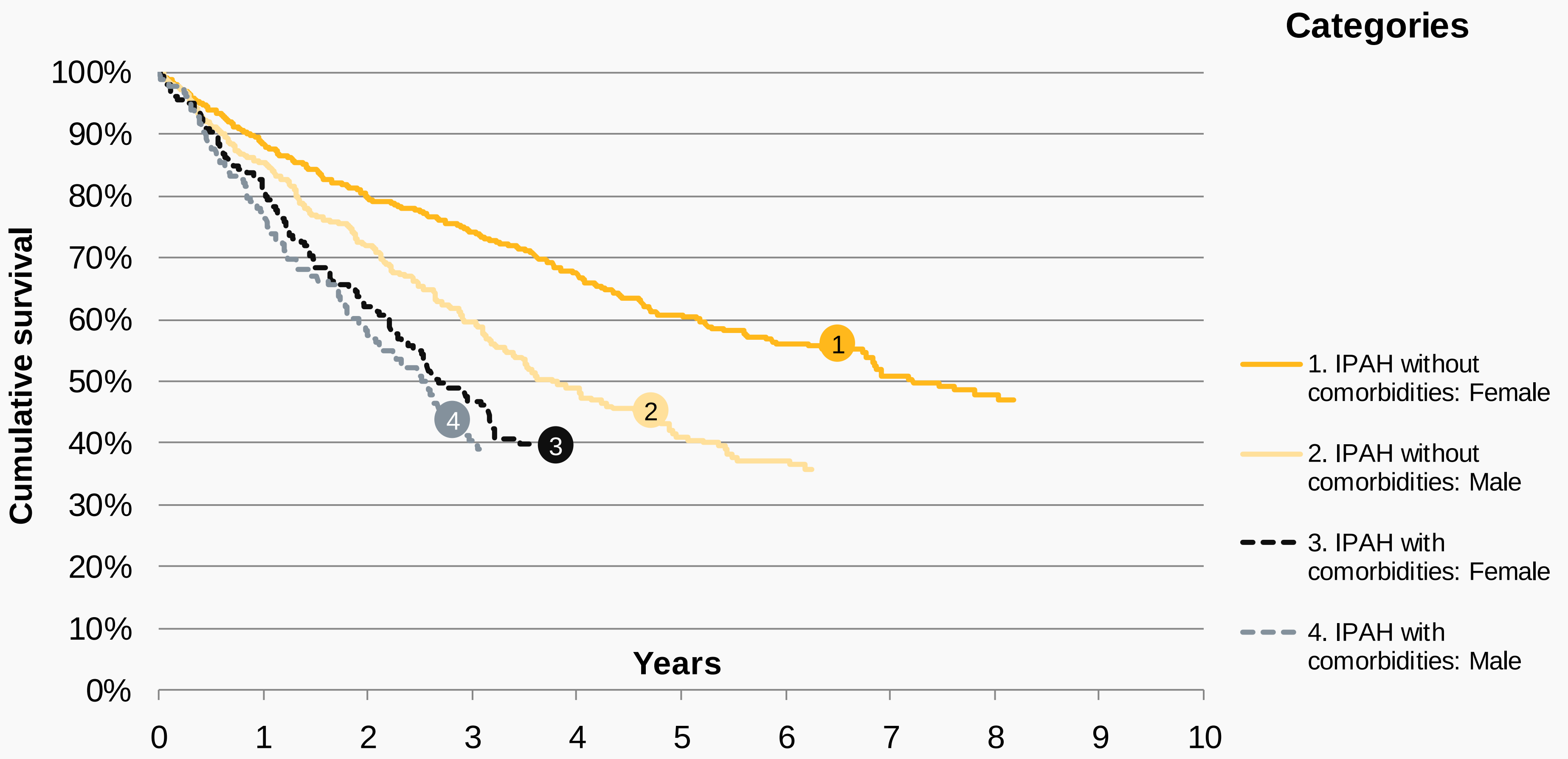
How is mortality data collected?

Date of death is submitted directly to NAPH by the PH Centres, then supplemented by mortality data from the Office for National Statistics (ONS).

Patients submitted by the Scottish PH Centre (Golden Jubilee) cannot be traced at ONS or the National Records of Scotland because they are pseudo-nymised in NAPH[2]. Scotland are not included in the lung disease or left heart disease survival curves due to difficulties in obtaining death data for those patients.

Survival curves: Sex

Chart SC 1: Survival curves from diagnosis by sex and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19



Notes: **1. IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. See the Survival analysis: Overview slide for details on which patients are included and when the survival curve is stopped.

Survival curves: Sex
Annual survival

Table SC 1: Post-diagnosis survival by sex and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Years Elapsed	IPAH without comorbidities				IPAH with comorbidities			
	Female		Male		Female		Male	
	Patients	% survival	Patients	% survival	Patients	% survival	Patients	% survival
0	740	100	392	100	204	100	181	100
1	581	88	299	85	152	81	133	77
2	458	80	217	72	100	62	89	57
3	377	74	160	60	64	47	52	40
4	292	67	115	49	–	–	–	–
5	208	61	80	41	–	–	–	–
6	149	56	57	37	–	–	–	–
7	99	51	–	–	–	–	–	–
8	57	48	–	–	–	–	–	–
9	–	–	–	–	–	–	–	–

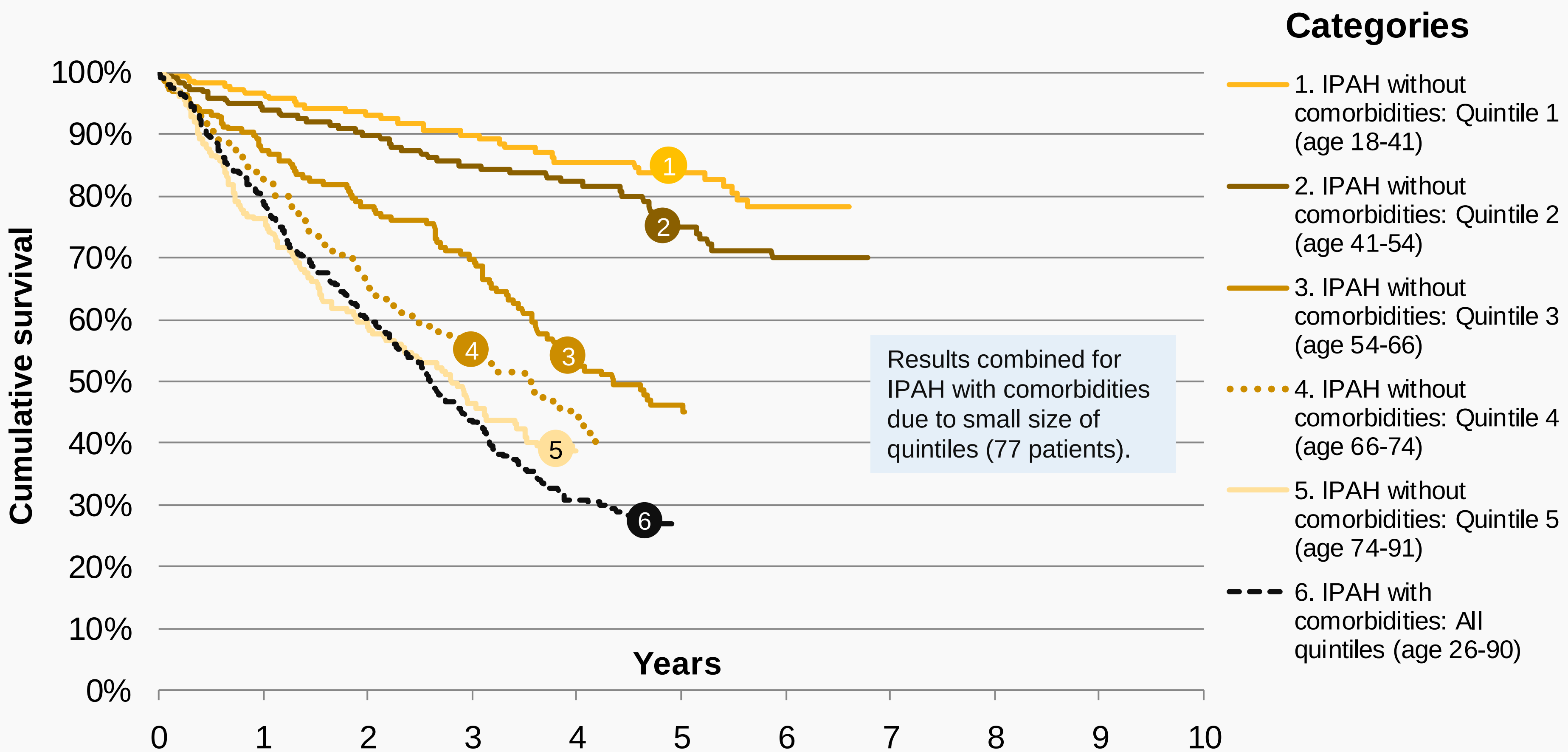
Breakdowns of the characteristics of each diagnostic group are available in the Group characteristics:

- Patient demographics (age, sex)
- PH characteristics including WHO functional class, six-minute walk test, quality of life (emPHasis-10 score) and haemodynamics (blood pressure)
- Lung function test results
- Therapy type (mono, combination or none)

Notes: 1. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. See the Survival analysis: Overview slide for details on which patients are included and when the survival curve is stopped.

Survival curves: Age at diagnosis

Chart SC 2: Survival curves from diagnosis by age at diagnosis and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19



Notes: **1**. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Results are combined for IPAH with comorbidities due to the small size of the quintiles (77 patients).
3. See the Survival analysis: Overview slide for details on which patients are included and when the survival curve is stopped.

Survival curves: Age at diagnosis
Annual survival

Table SC 2: Post-diagnosis survival by age at diagnosis and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Years Elapsed	IPAH without comorbidities										IPAH with comorbidities	
	Quintile 1 (age 18-41)		Quintile 2 (age 41-54)		Quintile 3 (age 54-66)		Quintile 4 (age 66-74)		Quintile 5 (age 74-91)		All quintiles (age 26-90)	
	Patients	% survival	Patients	% survival	Patients	% survival	Patients	% survival	Patients	% survival	Patients	% survival
0	227	100	227	100	226	100	226	100	226	100	385	100
1	192	97	191	94	171	87	170	83	156	76	285	79
2	161	93	158	90	134	78	120	66	102	59	189	60
3	133	90	137	85	107	70	90	55	70	46	116	43
4	108	86	114	82	73	54	62	44	–	–	67	31
5	88	84	85	75	51	46	–	–	–	–	–	–
6	59	78	63	70	–	–	–	–	–	–	–	–
7	–	–	–	–	–	–	–	–	–	–	–	–
8	–	–	–	–	–	–	–	–	–	–	–	–
9	–	–	–	–	–	–	–	–	–	–	–	–

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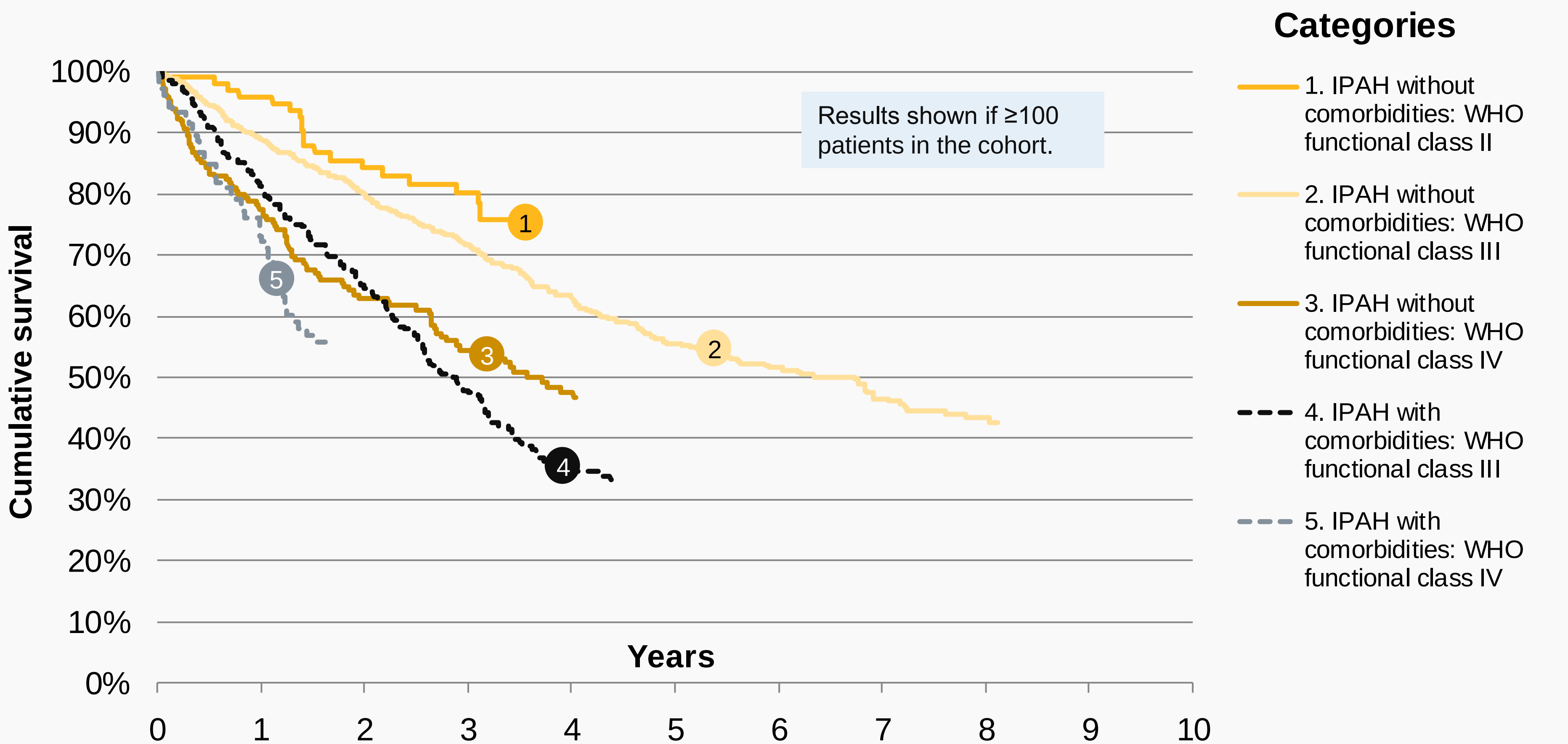
Survival curves: WHO functional class

Chart SC 3: Survival curves from diagnosis by WHO functional class and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19



Notes: **1**. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Results shown if ≥100 patients in the initial cohort.
3. See the Survival analysis: Overview slide for details on which patients are included and when the survival curve is stopped.

Survival curves: WHO functional class Annual survival

Table SC 3: Post-diagnosis survival by WHO functional class and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Years Elapsed	IPAH without comorbidities						IPAH with comorbidities			
	WHO functional class II		WHO functional class III		WHO functional class IV		WHO functional class III		WHO functional class IV	
	Patients	% survival	Patients	% survival	Patients	% survival	Patients	% survival	Patients	% survival
0	103	100	811	100	213	100	274	100	106	100
1	87	96	641	89	149	77	207	81	74	73
2	66	84	502	80	104	63	148	65	–	–
3	56	80	400	72	79	55	94	47	–	–
4	–	–	308	63	53	48	55	34	–	–
5	–	–	218	56	–	–	–	–	–	–
6	–	–	155	52	–	–	–	–	–	–
7	–	–	99	47	–	–	–	–	–	–
8	–	–	56	43	–	–	–	–	–	–
9	–	–	–	–	–	–	–	–	–	–

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- Therapy type (mono, combination or none)

Notes: **1. IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Results shown if ≥100 patients in the initial cohort.
3. See the Survival analysis: Overview slide for details on which patients are included and when the survival curve is stopped.

Group characteristics:

Demographics and PH characteristics

Group characteristics: Sex Demographics

Glossary

Table GC 1: Demographics by sex and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Group	Patients	Age at diagnosis				Days after diagnosis on which the cohort had 50 per cent mortality
	Number	Lower quartile	Median	Upper quartile	Mean	
IPAH without comorbidities						
…Female	740	42	57	71	56	7 years and 77 days
…Male	392	50	65	73	61	3 years and 279 days
IPAH with comorbidities						
…Female	204	61	69	76	67	2 years and 244 days
…Male	181	65	71	78	70	2 years and 211 days

Notes:
1. IPAH without comorbidities excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. See the Survival analysis: Overview slide for details on which patients are included and when the survival curve is stopped.

Group characteristics: Sex
PH characteristics

Glossary

Table GC 2: PH characteristics at diagnosis by sex and IPAH type,

Adult patients in the longitudinal cohort, Great Britain, 2009-19

Parameters	I = least severe IV = most severe				Further = healthier		0=best QoL 50=worst		Norm~14. 25+ =PH		Norm ~8		Norm= 0 to 7		Norm= ≤3		Norm= 65 to 75		Norm= 2.5 to 4.2	
Group	WHO functional class				Six-minute walk test (metres)		Quality of life: emPHasis-10 score (0-50)		Mean pulmonary artery pressure (PAP) (mm Hg)		Mean pulmonary wedge pressure (PWP) (mm Hg)		Mean right atrial pressure (RAP) (mm Hg)		Mean pulmonary vascular resistance (PVR) (Wood units)		Mixed venous oxygen saturation (SvO2) (%)		Cardiac index (L/min/m²)	
	I	II	III	IV																
	%	%	%	%	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
IPAH without comorbidities																		
…Female	1	10	71	19	247	151	29.1	13.5	52.3	13.1	9.7	4.5	10.0	5.8	12.6	6.3	62.3	10.0	2.2	0.8
…Male	0	8	74	18	230	152	30.1	12.9	49.2	11.8	10.1	4.8	9.4	5.4	10.3	4.6	62.3	9.6	2.2	0.7
IPAH with comorbidities																		
…Female	0	1	73	26	*	*	31.8	11.3	51.7	10.1	12.3	5.3	12.3	6.0	11.2	5.4	60.3	9.3	2.3	0.8
…Male	0	2	69	29	*	*	*	*	49.2	9.1	11.8	4.7	12.1	6.0	9.5	3.9	59.4	7.4	2.2	0.6

Notes: * = Insufficient values to produce robust calculation (<50 patients).
1. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. PH characteristics derived from the closest measurement within 90 days of diagnosis. See the Glossary: PH Characteristics slides 1 and 2 for further information. 3. SD = Standard deviation.
4. **Parameters** (top row of chart) for some characteristics include normal ranges (e.g. 0 to 7 for RAP), which show the typical range for a healthy person. 5. See the Survival analysis: Overview slide for details on which patients are included and when the survival curve is stopped.

Group characteristics: Sex
Lung function tests

Glossary

Table GC 3: Lung function tests at diagnosis by sex and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Group	Mean per cent predicted forced expiratory volume in one second (FEV1) (%)		Mean per cent predicted forced vital capacity (FVC) (%)		Mean per cent predicted transfer factor for carbon monoxide (TLco) (%)	
	Mean	SD	Mean	SD	Mean	SD
IPAH without comorbidities						
…Female	86	19	93	20	56	20
…Male	81	18	89	18	51	22
IPAH with comorbidities						
…Female	77	24	90	27	36	20
…Male	74	17	86	18	36	15

Notes:
1. IPAH with comorbidities is defined as: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Lung function test results derived from the closest measurement within 90 days of diagnosis. See the Glossary: Lung function tests slide for further details. **3**. SD = Standard deviation.
4. See the Survival analysis: Overview slide for details on which patients are included.

Group characteristics: Sex Therapy type

Glossary

Table GC 4: Latest recorded therapy type by sex and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Group	Patients	Mono-therapy		Dual therapy		Triple+ therapy		No therapy	
	Number	Patients	%	Patients	%	Patients	%	Patients	%
IPAH without comorbidities									
…Female	740	242	33	388	52	86	12	24	3
…Male	392	149	38	192	49	34	9	17	4
IPAH with comorbidities									
…Female	204	112	55	84	41	4	2	4	2
…Male	181	102	56	76	42	1	1	2	1

Notes:
1. **IPAH with comorbidities** is defined as: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Therapy type is derived from the number of concurrent drug classes prescribed on the latest recorded therapy date. See the Glossary: PH Characteristics (1) slide for details of how drug class is defined.
3. See the Survival analysis: Overview slide for details on which patients are included.

Group characteristics: Age at diagnosis Demographics

Table GC 5: Demographics by age at diagnosis and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Group	Patients	Age at diagnosis				Female	Days after diagnosis on which the cohort had 50 per cent mortality
	Number	Lower quartile	Median	Upper quartile	Mean	%	
IPAH without comorbidities							
…Quintile 1 (age 18-41)	227	27	32	36	31	76	*Not reached at 6 years and 218 days*
…Quintile 2 (age 41-54)	227	45	47	51	48	72	*Not reached at 6 years and 283 days*
…Quintile 3 (age 54-66)	226	58	60	63	60	64	4 years and 127 days
…Quintile 4 (age 66-74)	226	68	70	72	70	56	3 years and 203 days
…Quintile 5 (age 74-91)	226	76	78	82	79	59	2 years and 291 days
IPAH without comorbidities							
…All quintiles (age 26-90)	385	63	70	77	68	53	2 years and 218 days

Notes:
1. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Results are combined for IPAH with comorbidities due to the small size of the quintiles (77 patients).
3. See the Survival analysis: Overview slide for details on which patients are included and when the survival curve is stopped.

Group characteristics: Age at diagnosis – PH characteristics

Glossary

Table GC 6: PH characteristics at diagnosis by age and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Parameters	I = least severe IV = most severe				Further = healthier		0=best QoL 50=worst		Norm~14. 25+ =PH		Norm ~8		Norm= 0 to 7		Norm= ≤3		Norm= 65 to 75		Norm= 2.5 to 4.2	
Age group	WHO functional class: I	II	III	IV	Six-minute walk test (metres)		Quality of life: emPHasis -10 score (0-50)		Mean pulmonary artery pressure (PAP) (mm Hg)		Mean pulmonary wedge pressure (PWP) (mm Hg)		Mean right atrial pressure (RAP) (mm Hg)		Mean pulmonary vascular resistance (PVR) (Wood units)		Mixed venous oxygen saturation (SvO2) (%)		Cardiac index (L/min/m²)	
	%	%	%	%	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
IPAH without comorbidities																				
…Quintile 1 (18-41)	1	13	68	18	336	154	24.0	14.4	59.1	15.0	9.5	4.7	10.1	6.0	14.0	7.4	63.5	9.8	2.2	0.8
…Quintile 2 (41-54)	0	10	74	17	292	144	29.4	11.9	53.1	11.7	9.6	4.4	10.5	5.9	12.0	5.9	63.3	9.7	2.3	0.9
…Quintile 3 (54-66)	0	9	71	19	231	147	29.9	14.3	50.8	11.1	9.6	4.7	9.8	5.6	11.7	5.5	62.5	10.0	2.2	0.7
…Quintile 4 (66-74)	0	8	72	20	196	119	30.6	12.9	48.3	10.8	10.2	5.1	10.0	5.7	11.1	5.3	61.8	8.7	2.2	0.7
…Quintile 5 (74-91)	0	7	73	20	151	114	33.8	10.5	45.3	10.3	10.1	4.2	8.6	5.2	10.4	4.5	60.4	11.0	2.1	0.7
IPAH with comorbidities																				
…All quintiles (26-90)	0	1	71	28	191	142	32.8	11.5	50.5	9.7	12.1	5.1	12.2	6.0	10.4	4.8	59.8	8.5	2.2	0.7

Notes: 1. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Results are combined for IPAH with comorbidities due to the small size of the quintiles (77 patients).
3. PH characteristics derived from the closest measurement within 90 days of diagnosis. See the Glossary: PH Characteristics slides 1 and 2 for further information. **4.** SD = Standard deviation. **5.** **Parameters** (top row of chart) for some characteristics include normal ranges (e.g. 0 to 7 for RAP), which show the typical range for a healthy person.
6. See the Survival analysis: Overview slide for details on which patients are included and when the survival curve is stopped.

Group characteristics: Age at diagnosis – Lung function tests

Glossary

Table GC 7: Lung function tests at diagnosis by age and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Group	Mean per cent predicted forced expiratory volume in one second (FEV1) (%)		Mean per cent predicted forced vital capacity (FVC) (%)		Mean per cent predicted transfer factor for carbon monoxide (TLco) (%)	
	Mean	SD	Mean	SD	Mean	SD
IPAH without comorbidities						
…Quintile 1 (age 18-41)	85	16	90	16	64	18
…Quintile 2 (age 41-54)	84	21	91	21	57	20
…Quintile 3 (age 54-66)	86	18	94	19	50	22
…Quintile 4 (age 66-74)	84	21	92	22	44	18
…Quintile 5 (age 74-91)	*	*	*	*	*	*
IPAH with comorbidities						
…All quintiles (age 26-90)	76	22	88	24	36	18

Notes:
* = Insufficient values to produce robust calculation (<50 patients).
1. **IPAH with comorbidities** is defined as: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Results are combined for IPAH with comorbidities due to the small size of the quintiles (77 patients).
3. Lung function test results derived from the closest measurement within 90 days of diagnosis. See the Glossary: Lung function tests slide for further details. **4**. SD = Standard deviation.
5. See the Survival analysis: Overview slide for details on which patients are included.

Group characteristics: Age at diagnosis – Therapy type

Glossary

Table GC 8: Latest recorded therapy type by age at diagnosis and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Group	Patients	Mono-therapy		Dual therapy		Triple+ therapy		No therapy	
	Number	Patients	%	Patients	%	Patients	%	Patients	%
IPAH without comorbidities									
…Quintile 1 (age 18-41)	227	45	20	119	52	56	25	7	3
…Quintile 2 (age 41-54)	227	62	27	122	54	35	15	8	4
…Quintile 3 (age 54-66)	226	74	33	123	54	21	9	8	4
…Quintile 4 (age 66-74)	226	93	41	120	53	7	3	6	3
…Quintile 5 (age 74-91)	226	117	52	96	42	1	0	12	5
IPAH with comorbidities									
…All quintiles (age 26-90)	385	214	56	160	42	5	1	6	2

Notes:
1. IPAH with comorbidities is defined as: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Results are combined for IPAH with comorbidities due to the small size of the quintiles (77 patients).
3. Therapy type is derived from the number of concurrent drug classes prescribed on the latest recorded therapy date. See the Glossary: PH Characteristics (1) slide for details of how drug class is defined.
4. See the Survival analysis: Overview slide for details on which patients are included.

Group characteristics: WHO functional class Demographics

Table GC 9: Demographics by WHO functional class and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Group	Patients	Age at diagnosis				Female	Days after diagnosis on which the cohort had 50 per cent mortality
	Number	Lower quartile	Median	Upper quartile	Mean	%	
IPAH without comorbidities							
…WHO functional class II	103	40	54	69	54	70	*Not reached at 3 years and 208 days*
…WHO functional class III	811	45	60	72	58	64	6 years and 267 days
…WHO functional class IV	213	46	62	72	58	66	3 years and 259 days
IPAH with comorbidities							
…WHO functional class III	274	63	70	77	69	54	2 years and 311 days
…WHO functional class IV	106	63	70	75	68	50	*Not reached at 1 years and 224 days*

Notes:
1. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Results shown if ≥100 patients in the initial cohort.
3. See the Survival analysis: Overview slide for details on which patients are included and when the survival curve is stopped.

Group characteristics: WHO functional class – PH characteristics

Glossary

Table GC 10: PH characteristics at diagnosis by WHO functional class and IPAH type,

Adult patients in the longitudinal cohort, Great Britain, 2009-19

Parameters	Further = healthier		0=best QoL 50=worst		Norm~14. 25+ =PH		Norm ~8		Norm= 0 to 7		Norm= ≤3		Norm= 65 to 75		Norm= 2.5 to 4.2	
Group	Six-minute walk test (metres)		Quality of life: emPHasis-10 score (0-50)		Mean pulmonary artery pressure (PAP) (mm Hg)		Mean pulmonary wedge pressure (PWP) (mm Hg)		Mean right atrial pressure (RAP) (mm Hg)		Mean pulmonary vascular resistance (PVR) (Wood units)		Mixed venous oxygen saturation (SvO2) (%)		Cardiac index (L/min/m²)	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
IPAH without comorbidities																
…WHO functional class II	380	130	*	*	47.3	13.6	8.3	4.2	6.6	4.5	11.0	7.3	67.6	7.8	2.4	0.9
…WHO functional class III	245	137	28.4	13.0	51.0	12.7	10.0	4.6	9.6	5.2	11.4	5.5	62.8	9.5	2.2	0.7
…WHO functional class IV	133	134	37.9	10.5	54.0	11.7	9.9	4.8	12.2	6.8	13.8	6.1	58.2	10.7	2.0	0.8
IPAH with comorbidities																
…WHO functional class III	206	136	30.7	10.5	50.4	9.1	12.2	4.9	11.8	5.8	10.1	4.8	60.3	8.5	2.3	0.7
…WHO functional class IV	*	*	*	*	51.3	11.0	11.6	5.5	13.4	6.3	11.4	4.8	58.5	8.3	2.0	0.6

Notes: * = Insufficient values to produce robust calculation (<50 patients).
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Group characteristics: WHO functional class – Lung function tests

Glossary

Table GC 11: Lung function tests at diagnosis by WHO functional class and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Group	Mean per cent predicted forced expiratory volume in one second (FEV1) (%)		Mean per cent predicted forced vital capacity (FVC) (%)		Mean per cent predicted transfer factor for carbon monoxide (TLco) (%)	
	Mean	SD	Mean	SD	Mean	SD
IPAH without comorbidities						
…WHO functional class II	92	18	99	16	67	16
…WHO class III	85	18	92	19	55	21
…WHO class IV	80	20	86	21	48	23
IPAH with comorbidities						
…WHO functional class III	76	23	89	25	39	19
…WHO functional class IV	*	*	*	*	*	*

Notes:
* = Insufficient values to produce robust calculation (<50 patients).
1. **IPAH with comorbidities** is defined as: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Results shown if ≥100 patients in the initial cohort.
3. Lung function test results derived from the closest measurement within 90 days of diagnosis. See the Glossary: Lung function tests slide for further details. **4**. SD = Standard deviation.
5. See the Survival analysis: Overview slide for details on which patients are included.

Group characteristics: WHO functional class – Therapy type

Glossary

Table GC 12: Latest recorded therapy type by WHO functional class and IPAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Diagnostic group	Patients	Mono-therapy		Dual therapy		Triple+ therapy		No therapy	
	Number	Patients	%	Patients	%	Patients	%	Patients	%
IPAH without comorbidities									
…WHO functional class II	103	39	38	43	42	9	9	12	12
…WHO functional class III	811	285	35	418	52	83	10	25	3
…WHO functional class IV	213	64	30	117	55	28	13	4	2
IPAH with comorbidities									
…WHO functional class III	274	156	57	111	41	3	1	4	1
…WHO functional class IV	106	56	53	46	43	2	2	2	2

Notes:
1. **IPAH with comorbidities** is defined as: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Results shown if ≥100 patients in the initial cohort.
3. Therapy type is derived from the number of concurrent drug classes prescribed on the latest recorded therapy date. See the Glossary: PH Characteristics (1) slide for details of how drug class is defined.
4. See the Survival analysis: Overview slide for details on which patients are included.

Glossary

Glossary: Patient cohorts used in the report

The following patient cohorts are referred to in the report:

- **Managed patients** have a referral active at any point during the appropriate year.
- **Active patients** have a referral active on the 31 March of the appropriate year.
- **Active drug therapies** represent patients with at least one pulmonary hypertension drug therapy being prescribed on the 31 March of the appropriate year.
- The **cohort for longitudinal analysis** are patients whose first referral was on or after 1 April 2009. This cohort is used when there is a need to analyse data where the full treatment history for a patient is required. Since the Audit started on 1 April 2009, it is reasonable to expect that all of the events after that date will be recorded.

Scottish patients are submitted to the Audit by Golden Jubilee National Hospital using a pseudonymised patient identifier.

Therefore NHS Digital cannot identify a Scottish patient, although Golden Jubilee can. As a consequence:

a) It is not possible for NHS Digital to identify patients who have been treated in both Scotland and England. This means that a small number of patients may be included twice in the cohort for longitudinal analysis, once for the English recorded activity, and once for the Scottish.

b) It is not possible for the Office for National Statistics (ONS) to identify mortality data for Scottish patients. For English and Welsh patients, around one fifth of deaths are identified by ONS, not the Audit. It is therefore likely that some Scottish deaths will be missed by the Audit, though the proportion missing is not known.

Glossary: Acronyms and definitions (1)

Cardiac catheterization

An invasive diagnostic procedure that provides information about the structure and function of the heart. Cardiac catheterization is considered the gold standard method for confirming the presence, nature and severity of pulmonary hypertension.

CTEPH Chronic thromboembolic pulmonary hypertension

In chronic thromboembolic pulmonary hypertension (WHO Group 4) blood vessels are blocked or narrowed by the chronic consequences of blood clots, causing raised blood pressure in the pulmonary arteries.

NAPH National Audit of Pulmonary Hypertension

The National Audit of Pulmonary Hypertension measures the quality of care provided to people referred to pulmonary hypertension services in Great Britain.

PAH (APAH) Pulmonary arterial hypertension

In pulmonary arterial hypertension (WHO Group 1) the blood vessels become narrowed as the arterial walls stiffen and thicken, causing raised blood pressure in the pulmonary arteries. PAH resulting from another disease or cause such as connective tissues disease or congenital heart diseases is called associated-PAH or APAH.

PEA Pulmonary endarterectomy

This is a complex surgical procedure undertaken in selected patients with CTEPH, involving a heart-lung bypass and induced hypothermia where the circulation is slowed or stopped while blood clots and scar tissue are removed from the pulmonary arteries. The procedure can restore the raised blood pressure in the pulmonary circulation to normal in some patients with CTEPH.

Glossary: Acronyms and definitions (2)

PH **Pulmonary hypertension**

This is a medical condition in which the blood pressure is elevated in the blood vessels which supply the lungs. It can damage the heart and reduce its efficiency in supplying oxygen around the body.

PHA UK **Pulmonary Hypertension Association UK**

The only UK charity dedicated to people with pulmonary hypertension – it supports patients, works with healthcare professionals, provides research grants and lobbies for pulmonary hypertension.

phosphodiesterase 5 (PDE5) inhibitors

Phosphodiesterase 5 (PDE5) inhibitors are a type of targeted therapy used to treat some people with pulmonary hypertension. PDE5 inhibitors (i.e. sildenafil and tadalafil) cause the blood vessels to relax by stopping the PDE5 enzyme from working properly. This increases blood flow to the lungs and lowers blood pressure.

Significance

National results have been compared to those in the previous report to test whether changes over time are statistically significant at the 95% level ($p \leq 0.05$). This means that the result of the significance test has a 95% chance of being true[1] (see Glossary: Significance testing).

WHO **World Health Organization**

The World Health Organization is the United Nations coordinating authority on international health.

Notes: **1**. https://www.surveysystem.com/signif.htm.

Glossary: PH characteristics (1)

The **six-minute walking test** involves a person walking as far as possible up and down a corridor in six minutes. During the test, blood pressure, heart rate and oxygen levels are also measured. The test can provide clues to the cause of some symptoms of PH and is a good way to measure how effective PH treatment is, and if any changes to treatment are needed.

emPHasis-10 is a questionnaire which measures the impact of PH on a person's life, and how this changes over time. A total emPHasis-10 score is derived using simple aggregation of the 10 items. Scores range from zero to 50, with higher scores indicating worse quality of life as a result of the condition.

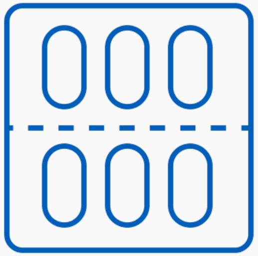
A patient's PH drug **therapy type** is defined as either:

- Mono-therapy
- Dual therapy
- Triple+ therapy
- No therapy

The therapy type is derived from the number of concurrent drug classes prescribed on the latest recorded therapy date.

The four **drug classes** are:

1) Phosphodiesterase 5 inhibitors;
2) Endothelin receptor antagonists;
3) Prostanoids;
4) Calcium channel blockers for vasoreactive PAH3.



Triple+ therapy also covers a small number of patients receiving all four drug classes concurrently.

Glossary: PH characteristics (2)

Mean pulmonary artery pressure (PAP) at rest is measured by cardiac catheterization. The normal mean pressure is 14 mm Hg and a person is considered to have PH when they have a mean baseline greater than or equal to 25 mm Hg[1].

Mean pulmonary wedge pressure (PWP) is an estimation of blood pressure on the left side of the heart (blood returning from the lungs to the heart), as measured by cardiac catheterization. The normal mean pressure is 8 mm Hg.

Mean right atrial pressure (RAP) is the blood pressure in the right atrium of the heart, as measured by cardiac catheterization, and is known to be affected by PH. A normal range is 0 to 7 mm Hg[1].

Mean pulmonary vascular resistance (PVR) is a measure of the resistance that must be overcome to push blood through the pulmonary circulatory system. A normal range is less than 3 Wood Units.

SvO2 (also known as mixed venous oxygen saturation) is the percentage of oxygen saturation in the pulmonary arterial blood, which is a measure of how much blood the heart is pumping round the body. The normal range is 65 to 75 per cent but it needs to be interpreted in the context of the clinical problem.

Cardiac index is a haemodynamic parameter that assesses how much blood a person's heart is pumping around the body. The cardiac index takes into account body size by dividing a person's cardiac output by their body surface area. The normal range for cardiac index is 2.5 to 4.2 L/min/m^2.

Notes: **1**. Galie N et al. Eur Heart J. 2016;37:67-119. Eur Resp J. 2015;46:903-75. https://erj.ersjournals.com/content/46/4/903.

Glossary: Lung function tests

A variety of lung function tests are used to help diagnose and monitor lung conditions such as PH.

Mean forced expiratory volume in one second (FEV1) measures how much air a person can exhale during the first second of a forced breath.

Mean forced vital capacity (FVC) measures how much air a person can exhale in one complete forced breath after taking a deep breath in.

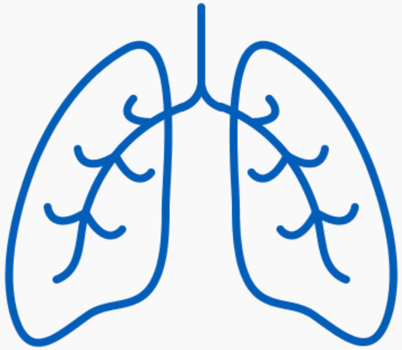
Mean transfer factor for carbon monoxide (TLco) measures how well a person's lungs are able to take up oxygen from the air they breathe and transfer it into the blood.

To become meaningful, results from lung function tests need to be adjusted for the patient's size and demographics. The equations in the table below use sex, age and height to produce predicted values for each patient. The measured results from each lung function test can then be expressed as a **per cent predicted** e.g. 50 per cent if the measured result is half the predicted result.

Equations to produce lung function test predictions[1]

Lung function test	Equation	
	Female A = 18 to 70 H = 1.45 to 1.80	Male A = 18 to 70 H = 1.55 to 1.95
FEV1	3.95H - 0.025A - 2.60	4.30H - 0.029A - 2.49
FVC	4.43H - 0.026A - 2.89	5.76H - 0.026A - 4.34
TLco	8.18H - 0.049A - 2.74	11.11H - 0.066A - 6.03

Key: A = Age in years. H = Height in metres



Notes: **1**. J.E. Cotes, D.J. Chinn, Ph.H. Quanjer, J. Roca, J.-C. Yernault European Respiratory Journal 1993 6: 41-52; DOI: 10.1183/09041950.041s1693.

Glossary: World Health Organization (WHO) Functional Class

The severity of pulmonary hypertension symptoms are recorded using the World Health Organization (WHO) functional class:

- **Class** I: Patients with pulmonary hypertension but without resulting limitation of physical activity. Ordinary physical activity does not cause undue dyspnoea or fatigue, chest pain or near syncope.
- **Class II**: Patients with pulmonary hypertension resulting in slight limitation of physical activity. They are comfortable at rest. Ordinary physical activity causes undue dyspnoea or fatigue, chest pain or near syncope.
- **Class III**: Patients with pulmonary hypertension resulting in marked limitation of physical activity. They are comfortable at rest. Less than ordinary activity causes undue dyspnoea or fatigue, chest pain or near syncope.
- **Class IV**: Patients with pulmonary hypertension with inability to carry out any physical activity without symptoms. These patients manifest signs of right heart failure. Dyspnoea and/or fatigue may even be present at rest. Discomfort is increased by any physical activity.

This is akin to New York Heart Association functional class. Efforts are made to record a WHO functional class for all patients at each hospital visit but this is not possible in all cases.

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